Task 1 — Basic Coding: Line-Position Estimation and Motor-Command Limiting

1. Objective

The task implements two essential functions for a line-following robot: estimating the line position using five IR sensors and limiting motor commands to a specified range. The C++ program also validates inputs and tests different edge cases.

2. Part A — Line-Position Estimation

The robot uses the weighted-centroid method to estimate the line's position relative to the centre of the sensor array.

Weighted Centre Formula:

$$Position = \Sigma(Reading\ X\ weight) \div \Sigma(Reading)$$

Sensor	Leftmost	Left	Centre	Right	Rightmost
Weight	−2	−1	0	+1	+2

The `estimateLinePosition` function checks the sensor count, validates readings within 0–4095, calculates the weighted sum and total reading, and determines the position. If all readings are zero, it returns failure to avoid division by zero.

Example:

Sensor readings	Weighted sum	Total reading	Position
0, 100, 300, 200, 0	100	600	+0.167

A positive position indicates that the detected signal is weighted towards the right, while a negative position indicates the left.

3. Part B — Motor-Command Limiting

The `clampMotorCommand` function restricts motor commands to a specified maximum magnitude while preserving their signs.

Clamped Reading = Max(-Maximum limit, Min(Maximum limit, Input))

Input command	Maximum limit	Output	Explanation
300	255	255	Positive command limited
−300	255	−255	Negative command limited
100	255	100	Within the permitted range
−50	0	0	Command restricted to zero
100	−1	Invalid	Negative limit rejected

The function returns a success or failure status and stores the result through a reference parameter. An additional test checks an extreme negative integer to verify that its command is limited correctly.

4. Testing and Conclusion

Function	Edge cases tested
Line-position estimation	All-zero readings, negative readings, out-of-range readings and incorrect sensor count
Motor-command limiting	Commands exceeding either limit, zero limit, negative limit and extreme negative command

The weighted-centroid function estimates the line's relative position, while the motor-command limiting function keeps commands within permitted boundaries. Together, they provide basic building blocks for a line-following robot, although the program does not directly control physical motors or implement a complete line-following controller.